Remaining-Points Check of the Updated Calculation

Check note

7 June 2026

Executive summary

This note checks the full active body of the updated TeX/PDF, not only the shortcut section. The no-shortcut front part mostly passes direct finite matrix checks: the preliminary identities, the no-shortcut first-passage formula, the killed propagator, and the Chebyshev compact forms agree with direct sums or absorbing-matrix resolvents to roundoff. The front part still has two local points worth adjusting: the compact Eq. (13) upper limit appears to use the long-mode range if read literally after f_k, α_k are defined, and two identities on line 365 are off by 1/2.

The shortcut part has a separate downstream sign-propagation point. The updated manuscript has corrected the shortcut resolvent sign through Eq. (31), but the correction has not been propagated consistently. The first main remaining shortcut point is Eq. (32): after summing the corrected Eq. (31), the denominator should remain

$$1 + z\beta(1-q)\widetilde{W}_u(u,z),$$

not $1 - z\beta(1-q)\widetilde{W}_u(u,z)$. Because Eq. (33), Eq. (40), Eq. (41), and the final tail paragraph are downstream of Eq. (32), they inherit this remaining sign mismatch.

The shortest way to see the point is:

$$\boxed{\text{correct Eq. (31)}} \Longrightarrow \boxed{\text{Eq. (32) denominator should be plus}}$$
$$\Longrightarrow \boxed{\text{Eq. (33) and Eq. (40) should have plus shortcut terms}}.$$

For the symmetric antipodal case, the safest corrected downstream expression is the closed form

$$\boxed{\widetilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}}$$

where $N = 2L$, $a = q/[\beta(1-q)]$, $y = 1 + (1/z - 1)/q$, and $U_{-1} = 0$. The time-domain tail is controlled by the roots of

$$D(y) = aT_L(y) + U_{L-1}(y),$$

not by a generic $t\alpha_1^t$ term.

This check covers the active manuscript body up to \end{document} in the updated TeX. Material after that point is commented-out scratch text and is not treated as part of the active PDF.

Check scope

The working instruction for this pass is:

> Check the latest 'Calculations.tex' and 'Calculations.pdf' as a full active manuscript, not only the shortcut section beginning around Eq. (29). Check the preliminary identities, no-shortcut

first-passage formulae, killed propagator formulae, compact modal notation, Chebyshev generating functions, and the special antipodal self Green function. For each formula, distinguish numerically verified formulae from notationally fragile formulae, formulae that differ if used literally, points that affect downstream conclusions, and compile/self-containedness points.

Pre-shortcut validation

The following finite checks were run before evaluating the shortcut section. They compare the printed formulae with direct finite sums, transition matrices, absorbing submatrices, and resolvents.

Check	Max difference	Interpretation
Eq. (1)–(4) preliminary identities	2.486900×10^{-14}	passes
Eq. (8) periodic propagator	5.551115×10^{-16}	passes
Eq. (9)–(12) first-passage time form	2.498002×10^{-16}	passes
Eq. (13), corrected compact upper $N/2$	2.775558×10^{-16}	passes
Eq. (13), literal printed upper $N-1$	4.811867×10^{-1}	differs
Eq. (16) killed propagator time form	1.642205×10^{-15}	passes
Eq. (19)–(20) killed propagator GF sums	7.993606×10^{-15}	passes
Eq. (21)–(22) Chebyshev modal sums	5.551115×10^{-15}	passes
Eq. (27)–(28) self Green functions	3.552714×10^{-15}	passes

Thus the front part is broadly consistent. Its main formula-level point is the compact-mode indexing in Eq. (13). Once f_k and α_k have been defined for $k = 1, \dots, N/2$, the compact generating-function sum must use $\sum_{k=1}^{N/2}$, not $\sum_{k=1}^{N-1}$. The Chebyshev expression on the right-hand side agrees with the corrected compact sum.

Core corrected chain

When the shortcut goes from u into the absorbing target v, it is killing inside the transient state space. The corrected first-passage generating function is

$$\boxed{\widetilde{F}_{n_0}(v,z) = \widetilde{\mathcal{F}}_{n_0}(v,z) + qz\,\widetilde{W}_{n_0}(u,z)\left[1 - \widetilde{\mathcal{F}}_u(v,z)\right]\widetilde{H}(z)} \tag{1}$$

with

$$\boxed{\widetilde{H}(z) = \frac{\beta(1-q)}{q}\,\frac{1}{1 + z\beta(1-q)\widetilde{W}_u(u,z)}.} \tag{2}$$

Thus the corrected version of Eq. (32) is

$$\boxed{\widetilde{F}_{n_0}(v,z) = \widetilde{\mathcal{F}}_{n_0}(v,z) + \frac{z\beta(1-q)\widetilde{W}_{n_0}(u,z)\left[1 - \widetilde{\mathcal{F}}_u(v,z)\right]}{1 + z\beta(1-q)\widetilde{W}_u(u,z)}.} \tag{3}$$

In time domain this gives, for $t \geq 1$,

$$\boxed{F_{n_0}(v,t) = \mathcal{F}_{n_0}(v,t) + q\sum_{t'=0}^{t-1} W_{n_0}(u,t-1-t')\sum_{t''=0}^{t'}\left[\delta_{t',t''} - \mathcal{F}_u(v,t'-t'')\right]H(t'').} \tag{4}$$

Corrected direct-route form

If one follows the original Eq. (41) route, $H(0)$ must be split off from the positive-time pole expansion:

$$h_0 = H(0) = \frac{\beta(1-q)}{q} = \frac{1}{a}, \qquad H(c) = \sum_k c_k s_k^{c-1} \quad (c \geq 1).$$

The needed convolution kernels are

$$A_t^{(+)}(x;s) = \frac{s^{t-1} - x^{t-1}}{s-x}, \qquad B_t^{(0)}(x,y) = \frac{x^{t-1} - y^{t-1}}{x-y},$$

$$B_t^{(+)}(x,y;s) = [z^{t-3}] \frac{1}{(1-zx)(1-zy)(1-zs)}.$$

Then the corrected long expression has four shortcut-convolution blocks:

$$\begin{aligned}
F_{n_0}(v,t) = &\sum_{\ell=1}^{L} f_\ell(n_0,v)\alpha_\ell^{t-1} \\
&+ qh_0 \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)\gamma_r^{t-1} + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)\alpha_r^{t-1} \right] \\
&+ q\sum_k c_k \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)A_t^{(+)}(\gamma_r;s_k) + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)A_t^{(+)}(\alpha_r;s_k) \right] \\
&- qh_0 \sum_{\ell=1}^{L} f_\ell(u,v) \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)B_t^{(0)}(\gamma_r,\alpha_\ell) + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)B_t^{(0)}(\alpha_r,\alpha_\ell) \right] \\
&- q\sum_k c_k \sum_{\ell=1}^{L} f_\ell(u,v) \left[\sum_{r=1}^{N-1} G_r^{(1)}(n_0,u,v)B_t^{(+)}(\gamma_r,\alpha_\ell;s_k) + \sum_{r=1}^{L} g_r^{(2)}(n_0,u,v)B_t^{(+)}(\alpha_r,\alpha_\ell;s_k) \right].
\end{aligned} \tag{5}$$

This direct-route expression is useful for checking the original derivation, but it should recompress to the closed form in the executive summary. In that closed form, for simple roots $D(y_k) = 0$, $s_k = 1 - q + qy_k$,

$$\boxed{F_\rho(t) = \sum_k B_{\rho k} s_k^{t-1}, \qquad B_{\rho k} = \frac{qN_\rho(y_k)}{D'(y_k)},}$$

where $N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)$.

Checklist

ID	Location	Current point	Suggested action	Type
A0	Eq. (13), line 251	The compact sum is printed as $\sum_{k=1}^{N-1}$ even though f_k and α_k have just been defined with odd-mode compact indexing $k = 1, \dots, N/2$. Used literally, this gives a different generating function from the Chebyshev RHS/direct check.	Use $\sum_{k=1}^{N/2}$ in the compact form. The Chebyshev expression on the right agrees with this corrected compact sum.	Formula-local
A1	Eq. (32), line 403	The denominator is still $1 - z\beta(1-q)\widetilde{W}_u(u, z)$.	Use Eq. (3); the denominator is $1 + z\beta(1-q)\widetilde{W}_u(u, z)$.	Main
A2	Eq. (33), line 408	The shortcut term is written with a minus sign, while the following definition of H has the plus denominator.	Use Eq. (1): the term is $+qz\,\widetilde{W}_{n_0}(u, z)[1-\widetilde{\mathcal{F}}_u(v, z)]\widetilde{H}(z)$.	Main
A3	Eq. (40), line 458	The time-convolution inherits the minus sign from the Eq. (33) sign choice.	Use Eq. (4); the convolution term is positive.	Main
A4	Eq. (41), lines 463–465	The long expression does not yet give the preferred corrected expression: the baseline targets u instead of v, the global signs inherit Eq. (40), the $g^{(2)}$ sector is paired with γ_r in places where it should be paired with α_r, and $H(0)$ is not separated.	Use the direct-route form (5), or preferably the closed form and partial fractions above.	Main
A5	Final paragraph, line 468	The conclusion that the tail is generically dominated by $t\alpha_1^t$ is an intermediate-mode conclusion, not a conclusion from the recompressed generating function.	The tail is $B_{\rho 1} s_1^{t-1}$ when the dominant root is simple; the finite-time double-peak question is separate from this tail statement.	Main
A6	Final paragraph, line 468	The proposed finite transition $s_1 = \gamma_1$ is not supported for finite positive β in the corrected denominator.	At $s = \gamma_1$, the $\phi(1/s)$ term is zero and the denominator is $q/[\beta(1-q)] > 0$, so this crossing is not a root condition.	Main
B1	Eq. (38)–(39), lines 428–451	The pole expansion of H is treated as if only positive-time poles matter. The residue formula and root count are fragile.	Keep $H(0) = \beta(1-q)/q$ separate. Write $c_k = qT_L(y_k)/D'(y_k)$ with $D(y) = aT_L(y) + U_{L-1}(y)$, and solve $D(y) = 0$ directly.	Structural

ID	Location	Current point	Suggested action	Type
B2	Figure caption, lines 436–438	The caption says $q > 1$ is required for a non-zero shortcut probability, while the model uses $0 < q < 1$; it also uses the pole figure in a way that could be read as determining the finite-time peak transition.	Change to $0 < q < 1$, $\beta > 0$. Treat the figure as an H-pole visualization only, not as a standalone double-peak-transition criterion.	Structural
B3	Line 441 and Eq. (39), line 448	The text says $k = 1, \dots, N/2 - 1$ in the denominator, while the previous caption says $M = N/2$; the coefficient formula is much less robust than a derivative-residue expression.	Use $D(y_k) = 0$, $s_k = 1 - q + qy_k$, and $c_k = qT_L(y_k)/D'(y_k)$. Avoid claiming $c_k = -\infty$ at an asymptote.	Structural
B4	Line 365	Two of the three displayed trigonometric identities are off by 1/2.	The correct identities are $\sum 1/(1 - \cos(\pi k/N)) = (N^2 - 1)/3$ and $\sum (-1)^k/(1 - \cos(\pi k/N)) = -(N^2 + 2)/6$.	Structural
B5	Line 283	The text says $Q_{n_0}(n, t) = 0$ at the absorbing boundary and as $t \to \infty$. Q was defined as the unrestricted periodic propagator, so this statement does not match Q as defined.	This statement should refer to W, the killed propagator.	Structural
B6	Eq. (25), line 288	The compact form has a dangling $G_k^{(2)}$ term without a displayed sum, while the later equivalent form uses the correct compact α_k sum.	Add the missing sum or replace this line with the later γ_k/α_k form.	Structural
B8	Line 401	The text says Eq. (30) is summed, but the needed special case is the simplified Eq. (31) with $m = v$.	Reference the simplified equation directly before deriving Eq. (32).	Structural
B9	Line 257	The first displayed sentence calls $\sum_{s=1}^{t} \mathcal{F}_{n_0}(m, s)$ a survival probability, but this quantity is the cumulative absorption probability up to time t.	Call the first quantity cumulative absorption; reserve survival probability for $R_{n_0}(t) = 1 - \sum_{s=0}^{t} \mathcal{F}_{n_0}(m, s)$.	Structural
C1	Line 1; lines 65–68	The attachment is not self-contained: direct compilation needs local style files and the figure. Also `idelinks` appears to be a typo and `customcolours` is loaded twice.	Either include the style files and `phifunction.pdf`, or replace them with portable definitions. Move link options to `hyperref`.	Compile
C2	Line 189	There is an empty label, \label{eq:}.	Replace it with a meaningful unique label or remove it.	Notation

ID	Location	Current point	Suggested action	Type
C3	Line 220	There is an extra closing parenthesis, and the condition says $n_0 \neq n$ where the context is the absorbing target.	Remove the extra parenthesis and clarify the condition as starting away from the absorbing site, e.g. $n_0 \neq m$, as appropriate.	Notation
C4	Lines 293–294	The arguments are written $n_0, n, , m$.	Remove the doubled comma.	Typo
C5	Line 370	The notation switches to $\widetilde{\mathcal{W}}$ while the rest of the manuscript uses $\widetilde{W}$.	Use a consistent killed-propagator symbol unless a distinct object is intended.	Notation
C6	Lines 438, 441, 451	Small text edits: `correspoding`, `dominator`, and `an$s_k`.	Correct to `corresponding`, `denominator`, and "and $s_k \neq \alpha_k$".	Typo
C7	Lines 171 and 433	Minor active-text copy points: "preliminaries identities" and the sentence "the denominator ... has M real zeros ... and are shown graphically".	Use "preliminary identities" or "preliminaries: identities", and rewrite the line 433 sentence so the subject agrees, e.g. "the zeros are shown".	Typo

Numerical checks

Shortcut sign benchmark

The direct benchmark is the finite stochastic shortcut matrix: starting from the lazy ring transition matrix, move $\lambda = \beta(1-q)$ from the $u \to u$ self-loop to the directed shortcut $u \to v$. Then compute the resolvent and the first-passage ratio

$$\widetilde{F}_{n_0}(v,z) = \frac{\widetilde{S}_{n_0,v}(z)}{\widetilde{S}_{v,v}(z)}.$$

Over the checked grid, the maximum differences were:

updated/original Eq. (32) sign vs stochastic matrix	3.669746×10^{-1}
corrected plus/plus Eq. (32) vs stochastic matrix	2.775558×10^{-16}
literal Eq. (29) ratio vs original Eq. (32)	1.318390×10^{-16}
stochastic-sign Eq. (29) ratio vs corrected Eq. (32)	5.551115×10^{-17}
Eq. (19) $W_{n_0}(u,z)$ vs absorbing matrix	1.043610×10^{-14}
Eq. (28) $W_u(u,z)$ vs absorbing matrix	1.687539×10^{-14}

This isolates the point to the shortcut sign propagation, not to the W inputs.

Identity check

For $N = 10$,

$$\sum_{k=1}^{N-1} \frac{1}{1-\cos(\pi k/N)} = 33, \qquad \frac{2N^2+1}{6} = 33.5,$$

and

$$\sum_{k=1}^{N-1} \frac{(-1)^k}{1-\cos(\pi k/N)} = -17, \qquad \frac{1-N^2}{6} = -16.5.$$

So the second and third identities displayed on line 365 are not exact.

Pre-shortcut check summary

The pre-shortcut checks show that the main no-shortcut formulae are numerically sound once the compact index in Eq. (13) is corrected:

preliminary identities Eq. (1)–(4)	2.486900×10^{-14}
first-passage Eq. (9)–(13), corrected compact index	2.775558×10^{-16}
killed propagator Eq. (16), Eq. (19)–(20)	7.993606×10^{-15}
Chebyshev forms Eq. (21)–(22), Eq. (27)–(28)	5.551115×10^{-15}

The literal printed Eq. (13) compact upper limit $N-1$, however, differs from the finite-matrix benchmark by up to 4.811867×10^{-1} on the check grid.

Shortest collaborator-facing statement

A concise way to describe the situation is:

> I checked the active manuscript body, including the formulae before the shortcut section. The no-shortcut first-passage and killed-propagator formulae agree with finite-matrix checks to roundoff once the compact Eq. (13) index is read as $k = 1, \ldots, N/2$. The earlier local points I see are therefore the printed compact upper limit in Eq. (13), the two line-365 identities, and some notation/text points.
>
> In the shortcut section, the updated resolvent sign is correct through the simplified absorbing case, but the first-passage formula still carries the previous denominator sign. Once the denominator in Eq. (32) is corrected, Eq. (33) and the time-convolution Eq. (40) also need the plus shortcut term. The long Eq. (41) should be treated with care as a tail conclusion in its present form: after the $H(0)$ term is separated and the modes are recombined, the symmetric case reduces to the closed form with denominator $aT_L(y) + U_{L-1}(y)$, and the tail is controlled by the dominant root of that denominator.